Rational Number Project

Initial Fraction Ideas Lesson 12: Overview	Materials
Students are introduced to chips as a fraction model. They learn to represent a given fraction using different sets of chips as a unit.	• Chips or tiles with a different color on each side • Paper strips for folding • Student Page A

Teaching Actions

Warm Up

Which fraction is larger? Share your answer and rationale to a partner. What mental images for fractions did you picture to find the larger fraction?

$\frac{3}{4}$ $\frac{1}{2}$

Large Group Introduction

1. You use chips while students use paper folding to show the same fraction.

2. Model the fraction $\frac{2}{3}$ with the display chips in a way for all students to see.

   Say: I have 6 chips (show 6 chips with the white side up). This is my unit or whole that I am going to divided into equal groups. I'm going to partition them into 3 equal groups. Can you partition your paper strip into 3 equal parts?

   Ask: How are the displays alike and different?



3. Turn over 2 of the 3 equal groups of chips to show the tan side of the chips. Ask children to model this action on their paper strips by shading 2 of 3 equal parts.

   Say: I made 2 of 3 equal-sized groups tan.

Comments

This will be a challenging lesson for students. You may want to use two class periods to cover this material.

To reinforce important fraction concepts students are introduced to a new model for fractions (chips) by relating this model to a previous one (paper folding). This is a translation from one physical model to another. Seeing similarities between models helps students abstract important concepts.

Common error: Students model $\frac{2}{3}$ by making **groups of three** instead of **three equal groups**.

We use white square tiles found at any tile store as they are an inexpensive way to provide all students with their own chips.

Teaching Actions

Ask: What fraction of the chips is tan?

Ask: How are the chips and paper folding models alike? Different?



4. Summarize what you did by writing on the board.

   I started with a unit of _____ white chips.

   I divided the unit into _____ equal groups.

   I made _____ groups tan.

   _____ of _____ equal-sized groups are tan.

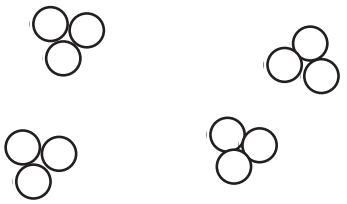
   What fraction of the groups is tan? ($\frac{2}{3}$)

5. Ask students to verbalize what they did with paper in a similar way. Conclude that there are many different models to show fractions.

6. Model other fractions using chips. Students should have their own chips. For the sake of consistency, use the white side to show the unit and use the tan side to show "amount shaded."

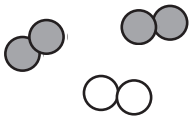
7. SAY**:** I have 12 chips. I want to show $\frac{1}{4}$ using these chips as my unit.

   ASK**:** How many equal-sized groups will I need? (four). Now divide the 12 chips into 4 equal groups.

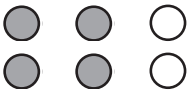


Comments

Students often need to exaggerate the grouping of the chips to highlight the separate groups.

Ex: $\frac{2}{3}$ 

They will spread out the sets and arrange the chips to touch to show a group. At this point don't use the array-model.

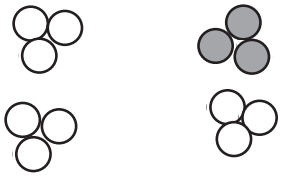


Some students say that 4/6 is covered. Informal discussion of equivalence is encouraged, but don't rush the idea.

Student verbalization is important. You might ask them to write a description for showing a fraction with chips.

Teaching Actions

ASK: To show $\frac{1}{4}$, how many equal groups must I make tan? (one)



8. Repeat for several more fractions.

Fraction	Unit
$\frac{2}{3}$	6
$\frac{1}{2}$	8
$\frac{1}{3}$	12
$\frac{3}{4}$	8

Small Group/Partner Work

9. Student Page A provides independent practice.

Wrap Up

10. Present these two stories to students. Ask them to decide which model, paper folding or chips, would be best to show the fraction the story.

    Story 1: Devan had 15 m & m's. She shared them equally between herself and 2 friends. What fraction of candy did she get?

    Story 2: Lianna has a Nestle Crunch bar. She plans to share it between herself and 2 friends. What fraction of the candy bar will each get?

Comments

Students will struggle a bit. As you move from group to group you may want to model the fraction using paper folding, going through each step as the students use the chips to copy your steps.

Students confuse the number of groups with the number of chips in each group so emphasize that $\frac{3}{4}$ means 3 out of 4 equal-sized groups. It does not mean to put 4 chips into a group.

Translations:

- Manipulative to manipulative to verbal; symbols to manipulative to verbal

Which fraction is larger? Share your answer and rationale to a partner.

What mental images for fractions did you picture to find the larger fraction?

$$\frac{3}{4} \quad \frac{1}{2}$$

Modeling Fractions with Chips

1. Show 3-fourths with chips. Use 8 chips in all. Draw a picture of your display.

2. Show 1-fourth with chips. Use 12 chips in all. Draw a picture of your display.

3. Show 2-sixths with chips. Use 12 chips in all. Draw a picture of your display.

4. Show 2-sixths with chips. Use 6 chips in all. Draw a picture of your display.

5. Show 1-fifth with chips. Use 15 chips in all. Draw a picture of your display.

6. On the back of this page, describe steps you would take to show $\frac{2}{3}$ using 12 chips.